

Euclid's Elements — Book V

Proposition 11

Formal Reconstruction — Technical Documentation

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CGJTEAMLAB

Ratios equal to the same ratio are equal to one another.

1 Introduction

Proposition V.11 is structurally one of the simplest propositions in the present Book V reconstruction. Its statement is the transitivity of equality of ratios:

$$a : b = c : d, \quad c : d = e : f \quad \implies \quad a : b = e : f.$$

In the Eudoxean interface adopted by the project, equality of ratios is defined pointwise by preservation of all three comparisons between positive integer multiples. Consequently V.11 becomes direct composition of equivalences.

The production file is

```
CGJteamLab/Proposition5_11.lean
```

and it imports only

```
CGJteamLab.HilbertInterfaceV
```

from the Book V infrastructure.

2 Euclid's Statement

The proposition is represented in the production file by the statement:

Ratios equal to the same ratio are equal to one another.

Formally, if

$$a : b = c : d$$

and

$$c : d = e : f,$$

then

$$a : b = e : f.$$

No strict ratio comparison is involved.

3 Eudoxean Equality of Ratios

The underlying relation is

HilbertEudoxusProportion Geo a b c d

which says that for every $m, n \in \mathbb{N}$, using the project's shifted positive-multiple indexing, the three comparisons

$$M_m(a) < M_n(b),$$

$$M_m(a) = M_n(b),$$

and

$$M_n(b) < M_m(a)$$

are respectively equivalent to

$$M_m(c) < M_n(d),$$

$$M_m(c) = M_n(d),$$

and

$$M_n(d) < M_m(c).$$

Thus equality of ratios is literally a family of three equivalences indexed by all positive finite multiples.

4 Why Transitivity Is Immediate

Assume

$$a : b = c : d$$

and

$$c : d = e : f.$$

Fix arbitrary m, n .

For the less-than comparison we have

$$M_m(a) < M_n(b) \iff M_m(c) < M_n(d)$$

from the first proportion, and

$$M_m(c) < M_n(d) \iff M_m(e) < M_n(f)$$

from the second.

Composing the two equivalences gives

$$M_m(a) < M_n(b) \iff M_m(e) < M_n(f).$$

Exactly the same argument applies to equality:

$$M_m(a) = M_n(b) \iff M_m(e) = M_n(f),$$

and to the reverse strict comparison:

$$M_n(b) < M_m(a) \iff M_n(f) < M_m(e).$$

Since m, n were arbitrary, the three clauses are precisely Definition 5 for

$$a : b = e : f.$$

5 Formal Statement

The production theorem is:

```
theorem euclid_proposition_5_11
  [HilbertIncidence Geo]
  [HilbertCongruence Geo]
  (a b c d e f : HilbertPositiveSegmentClass Geo)
```

```

(h1 : HilbertEudoxusProportion Geo a b c d)
(h2 : HilbertEudoxusProportion Geo c d e f) :
HilbertEudoxusProportion Geo a b e f := by

exact
  hilbertEudoxusProportion_trans
    Geo a b c d e f h1 h2

```

The Euclid-numbered proposition is intentionally a one-step wrapper around the general interface theorem

```

hilbertEudoxusProportion_trans

```

already proved in `HilbertInterfaceV.lean`.

6 The Interface Theorem

The reusable theorem has the same mathematical content:

```

hilbertEudoxusProportion_trans

```

Its proof introduces arbitrary m, n , obtains the three comparison equivalences from each hypothesis, and composes them with

```

Iff.trans

```

for each comparison type.

There is no Archimedean step, no order trichotomy, and no construction of a greater-ratio witness.

7 Lean Proof Architecture

The complete formal DAG is

```

h1 : HilbertEudoxusProportion Geo a b c d
h2 : HilbertEudoxusProportion Geo c d e f
      ↓↓
    ∀m,n, three Iff clauses from h1
    ∀m,n, three Iff clauses from h2
      ↓↓
    Iff.trans on each clause
      ↓↓
HilbertEudoxusProportion Geo a b e f
      ↓↓
euclid_proposition_5_11.

```

The proposition therefore contains no proposition-local helper lemmas.

8 Axiomatic Strength

The theorem requires only

```
[HilbertIncidence Geo]
[HilbertCongruence Geo]
```

because these structures are already needed to form positive segment classes and their comparison relation.

In particular, V.11 does not require

```
HilbertArchimedeanPlane
```

and does not use continuity.

This is in sharp contrast with V.8, where the Archimedean principle is essential to manufacture a strict ratio witness.

9 No New Magnitude Construction

V.11 introduces no new magnitude, no positive remainder, and no finite multiple chosen by an existence theorem.

All objects

$$a, b, c, d, e, f$$

are supplied as hypotheses, and the conclusion is obtained solely by the logical structure of Definition 5.

Thus the proposition is an especially clear example of a theorem that becomes short after the correct reusable notion has been isolated.

10 Relationship with Definition 5

The brevity of the formal proof is not accidental. It reflects the design of

```
HilbertEudoxusProportion
```

itself.

Definition 5 was encoded as preservation of all three possible comparisons for all positive multiple pairs. Such a relation is automatically transitive because equivalence is transitive.

Accordingly, V.11 is not an additional arithmetic property imposed on ratios. It is a structural theorem of the chosen Eudoxean equality relation.

11 Relationship with Proposition V.16

V.11 is used directly in the formal proof of V.16 alternando.

The V.16 proof first uses the common-multiple content of V.15 to obtain

$$a : b = M_m(a) : M_m(b)$$

and

$$c : d = M_n(c) : M_n(d).$$

Starting from

$$a : b = c : d,$$

V.11 then composes these equalities of ratios to produce

$$M_m(a) : M_m(b) = M_n(c) : M_n(d).$$

V.14 can then transport the comparison between the first terms to the second terms. Since m, n are arbitrary, this yields the alternated proportion.

Hence the relevant dependency chain is

$$\text{V.15} \longrightarrow \text{V.11} \longrightarrow \text{V.14} \longrightarrow \text{V.16}.$$

12 Classical DAG versus Formal DAG

At the mathematical level:

$$\begin{array}{c} a : b = c : d \\ c : d = e : f \\ \Downarrow \\ a : b = e : f. \end{array}$$

At the Definition 5 level:

$$\begin{array}{c}
\text{comparison of multiples for } a : b \\
\Downarrow \\
\text{comparison of multiples for } c : d \\
\Downarrow \\
\text{comparison of multiples for } e : f \\
\Downarrow \\
\text{comparison of multiples for } a : b \Downarrow \text{comparison of multiples for } e : f.
\end{array}$$

At the Lean level this is simply three uses of `Iff.trans` under universal quantification.

13 Structural Lesson

V.11 illustrates an important formalization principle.

A proposition can be historically central and still have a very short Lean proof when its mathematical content has already been placed at the correct abstraction level.

Here the work lies in defining Eudoxean equality of ratios correctly. Once that definition is available, transitivity is immediate.

The Euclid-numbered theorem should therefore remain a thin wrapper rather than duplicating the general proof in a proposition-specific file.

14 Conclusion

Proposition V.11 is the transitivity theorem for Eudoxean equality of ratios.

In the present architecture it is exactly the reusable theorem

`hilbertEudoxusProportion_trans,`

exposed under Euclid’s proposition number.

The proof requires no Archimedean reasoning and no construction. It consists only of composing the three comparison equivalences that constitute Definition 5. Its main later role is in V.16, where it composes common-multiple proportions with an existing ratio equality before V.14 performs the decisive comparison transfer.